Optimizing network measurements through self-adaptive sampling

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***Abstract*—Traffic sampling techniques are crucial and extensively used to assist network management tasks. Nevertheless, combining accurate network parameters' estimation and flexible lightweight measurements is an open challenge.**

In this context, this paper proposes a self-adaptive sampling technique, based on linear prediction, which allows to reduce significantly the measurement overhead, while assuring that sampled traffic reflects the statistical characteristics of the global traffic under analysis. The technique is multiadaptive as several parameters are considered in the dynamic configuration of the traffic selection process.

The devised test scenarios aim at exploring the proposed sampling technique ability to join accurate network estimates to reduced overhead, using throughput as reference parameter. The evaluation results, obtained resorting to real traffic traces representing wired and wireless aggregated traffic scenarios and actual network services, prove that the simplicity, flexibility and self-adaptability of this technique can be successfully explored to improve network measurements efficiency over distinct traffic conditions. For optimization purposes, this paper also includes a study of the impact of varying the order of prediction, i.e., of considering different degrees of past memory in the self-adaptive estimation mechanism. The significance of the obtained results is demonstrated through statistical benchmarking.

I. INTRODUCTION

Network measurements play a key role in the support of multiple aspects related to network management and engineering. Today's networks pose additional challenges to the design of efficient measurement strategies, as they have to deal with massive traffic volumes and simultaneously provide accurate feedback of the status of the network and supported services. For achieving this goal, traffic sampling techniques are broadly deployed in strategic network nodes, the measurement points. Their main objective is to select a subset of packets which will be used to estimate network parameters correctly, avoiding processing all network traffic [1]. The efficiency of a sampling technique may, therefore, be evaluated weighting the estimation accuracy and the measurement overhead.

Facing the drawbacks of commonly used deterministic and random sampling techniques (see discussion in Section II), adaptive sampling techniques have been proposed. In these techniques, the packet selection process is more flexible and self-adaptive, and its configuration may change dynamically during the measurement period, according to the value of an evaluated reference parameter. Despite the evolution of adaptive techniques in estimation accuracy, their main target is not focused on reducing the overhead associated with the volume of data involved in the sampling process. This aspect impacts directly on measurement costs and efficiency and motivates the definition of new proposals.

In this context, this paper presents a new self-adaptive traffic sampling technique, based on linear prediction, which aims to reduce the network measurement overhead without compromising estimation accuracy. For this purpose, the traffic selection process considers the levels of network activity, and is configured to reduce measurement impact when the network activity increases and measurement processes tend to be heavier for measurement points. This is achieved through a multiadaptive traffic selection functionality able to change the interval between samples and the sample size according to the network observed activity. Controlling these two variables and defining appropriate bounds for its values, this technique reduces significantly the overhead associated with the traffic sampling process while maintaining the accuracy in estimating the statistical network behavior when taking throughput as reference parameter.

This paper provides an encompassing proof-of-concept, exploring the proposed technique performance under distinct traffic scenarios. Based on real traffic traces, the measurement overhead, the throughput estimation accuracy, the impact of the order of prediction and the impact of the network activity on the measurement efficiency are evaluated. To demonstrate the significance of the obtained results, a statistical benchmarking is provided.

The remaining of this document is organized as follows: representative sampling approaches are debated in Section II; the multiadaptive sampling technique is described in Section III; the proof-of-concept objectives and methodology is presented in Section IV; the evaluation results are discussed in Section V; finally, the conclusions are drawn in Section VI.

II. RELATED WORK

Sampling techniques are usually characterized by the method adopted to select the packets that will integrate a traffic sample.

Systematic sampling techniques use a deterministic packet selection function based on (i) the packet position (count-based); (ii) the packet arrival time to the measurement point (time-base); or (iii) the packet contents (content-based) [2].

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Although this technique is simple, the traffic pattern resulting from deterministic sampling may still be heavy to measurement points and produce biased samples.

Random sampling techniques aim to tackle bias samples resorting to a random function to select packets [2]. Nevertheless, these techniques cannot be deployed to estimate multipoint metrics, such as edge-to-edge delay, since the sampling processes on the two measurement points involved are not correlated and there are no guarantees that samples will consist of the same packets [3].

Adaptive sampling techniques consider the value of a reference parameter observed during the measurement period, and the packet selection process may change dynamically according to this value. These techniques are generally developed for a specific estimation parameter, such as packet loss [4] [5] or delay and jitter [6] [7]. A larger group of proposals are aimed at traffic characterization and SLA monitoring [8] [9] [10] [7].

Adaptive techniques are usually based on Fuzzy Logic and Linear Prediction. In adaptive sampling based on fuzzy logic [11] [12], a controller adjusts the sampling rate based on past similar experiences, determining the most appropriate action for current traffic conditions [13]. This approach requires a long-term database to store the knowledge and the possible action for each situation. Linear prediction based techniques [14] [15] try to forecast network behavior based on an observed parameter in past samples. In these techniques, when the prediction is correct, the sampling rate can be reduced, while inaccurate predictions indicate a change in network activity and, therefore, an increase in sampling rate is required to determine the new pattern behavior [13]. In this sampling approach only a fixed number of samples is stored in the measurement point which are then used in the prediction process. However, if the sampling frequency increases more resources will be required from the measurement point, precisely at the most critical moment of its operation.

The sampling technique proposed in this paper attempts to decrease the resource consumption of performing measurements during high network activity periods, while maintaining the accuracy of network statistical behavior estimation.

III. Multiadaptive Sampling Technique

A. Concepts

This paper considers the following definitions:

- *Sample:* subset of the network packets that are selected at the measurement point and are considered in the estimation of network parameters. These packets are also used by the adaptive measurement algorithms as input for the estimation of the reference parameter;
- *Sample size:* time interval in which all incoming packets at the measurement point are selected and captured to compose a sample unit;
- *Interval between samples:* time interval in which all incoming packets are ignored for measurement purposes. During this time, the behavior of the network is not considered for the estimation of parameters and for the configuration of the adaptive sampling rate;
- *Reference parameter:* observed value in each sample, e.g. throughput, used as measurement parameter and input for configuring the sampling adaptation parameters.

B. Rational

The multiadaptive sampling technique proposed in this work improves the sampling reactivity and efficiency, considering both the interval between samples and the sample size as adjustable parameters. For this purpose, the present proposal uses a solution based on linear prediction to identify changes in the network activity.

The usage of linear prediction was presented previously in [13], claiming as advantages the simplicity of implementation and low consumption of resources. However this approach only considers the interval between samples as adjustable parameter. As consequence, the sampling frequency is increased whenever a noticeable increase in network activity occurs in order to allow detecting conveniently new traffic. This increasing in frequency sampling implies a higher consumption of processing and storage resources at the most critical periods of its operation. Hence, the sample size should be reduced to mitigate overhead while maintaining the representativeness of samples in the estimation of the parameters under analysis.

To meet this requirement, we propose a set of rules for varying the sample size according to the level of network activity identified by linear prediction. These rules increase and decrease the sample size differently depending on the observed network behavior. This dynamics is expressed in Algorithm 1 and described below. Furthermore, the present proposal only considers packets belonging to previously collected samples to drive future sampling decisions, clearly reducing the processing overhead in [13] that considers all traffic, regardless of whether or not the packets belong to a sample. In addiction, to ensure that the adaptive parameters are properly bounded, allowing to guarantee representative data for capturing network behavior, thresholds are set experimentally and presented in the following section.

C. Multiadaptive Sampling Technique Description

The multiadaptive technique uses linear prediction to estimate the future value of the reference parameter, which is then used to determine the next interval between samples and the size of the next sample.

Algorithm 1 illustrates the decision process regarding the next interval between samples ΔT_{next} and next sample size ΔS_{next}. Algorithm 2 presents the procedure used to calculate the expected value of the reference parameter Xp for the next collected sample, taking into account the last N values of this parameter. N determines the *order* of prediction of the sampler, setting it as more or less reactive to the past network behavior.

In Algorithm 1, the first samples are collected according to the sampler order (lines 2-7). The size of these samples is previously defined and will be used as an input of Algorithm 2 (line 11). When a new sample is collected (line 12), the reference parameter S is estimated (line 13) and is compared

input : ΔT_{actual} - Initial interval between samples
ΔS_{actual} - Initial sample size
output: Sampled traffic

1 **begin**
2 **for** $i \leftarrow 1$ **to** $order$ **do**
3 */* capture a new sample */*;
4 newSample(ΔT_{actual}, ΔS_{actual});
5 */* $X[i]$ stores the reference parameter of the last sample */*;
6 $X[i] \leftarrow$ referenceParameter();
7 $T[i] \leftarrow \Delta T_{actual}$; */* kept unchanged */*
8 **end**
9 **repeat**
10 */* Predictor() corresponds to Algorithm 2 */*;
11 $X_p \leftarrow$ Predictor(X, ΔT, T);
12 newSample ($\Delta T_{actual}, \Delta S_{actual}$);
13 $S \leftarrow$ referenceParameter();
14 $m \leftarrow abs((X_p - X[order])/(S - X[order]))$;
15 **case**$[m < m_{min}]$ */* underestimation */*
16 $\Delta T_{next} \leftarrow m * \Delta T_{actual}$;
17 $\Delta S_{next} \leftarrow \Delta S_{actual} - (\Delta S_{actual}/K)$; K=4
18 **case**$[m_{min} \leq m \leq m_{max}]$ */* correct */*
19 $\Delta T_{next} \leftarrow \Delta T_{actual}$;
20 $\Delta S_{next} \leftarrow \Delta S_{actual}$;
21 **case**$[m_{max} < m]$ */* overestimation */*
22 $\Delta T_{next} \leftarrow \Delta T_{actual} + 1sec$;
23 $\Delta S_{next} \leftarrow \Delta S_{actual} + (\Delta S_{actual}/K)$; K=10
24 **case**$[m = undefined]$ */* unchanged */*
25 $\Delta T_{next} \leftarrow \Delta T_{actual} * 2$;
26 $\Delta S_{next} \leftarrow \Delta S_{actual} + (\Delta S_{actual}/K)$; K=10
27 */* sampling frequency bounds (sec) */*
28 **if** $\Delta T_{next} < 0.1$ **then**
29 $\Delta T_{next} \leftarrow 0.1$;
30 **end**
31 **if** $\Delta T_{next} > 8$ **then**
32 $\Delta T_{next} \leftarrow 8$;
33 **end**
34 */* sampling size bounds (sec) */*
35 **if** $\Delta S_{next} < 0.1$ **then**
36 $\Delta S_{next} \leftarrow 0.1$;
37 **end**
38 **if** $\Delta S_{next} > 2$ **then**
39 $\Delta S_{next} \leftarrow 2$;
40 **end**
41 $\Delta T_{actual} \leftarrow \Delta T_{next}$;
42 $\Delta S_{actual} \leftarrow \Delta S_{next}$;
43 */* update vectors T and X */*
44 updateVectors($\Delta T_{actual}, S$);
45 **until** *endOfSampling*;
46 **end**

Algorithm 1: Multiadaptive technique main program

input : X - Reference parameter vector
ΔT - Actual interval between samples
T - Times vector
output: forecast - Reference parameter prediction

for $i \leftarrow 1$ **to** $order - 1$ **do**
$sum \leftarrow sum + abs((X[i+1] - X[i])/T[i])$;
end
$forecast \leftarrow X[order] + ((\Delta T/(order - 1)) * sum$;
return ($forecast$);

Algorithm 2: Reference parameter predictor

with the expected value Xp (line 14). This comparison leads to a factor m, used to determine the activity level of the network. Note that, m is close to 1 when the expected value Xp is close to the current value of S. The range of values that satisfies this condition is defined as $m_{min} < 1 < m_{max}$. The values of m_{min} and m_{max} are set experimentally as $m_{min} = 0.9$ and $m_{max} = 1.1$ due to its good performance for multiple traffic types.

If $m < m_{min}$ the reference parameter is changing faster than predicted. This behavior indicates more network activity than expected. Thus, the interval between samples should be decreased to achieve more accurate values, which will be used in following predictions. However, the increase in the sampling frequency leads to collecting more data in the sampling process. To avoid this increase in data volume, the sample size is decreased by a factor $1/K$, with $K = 4$ (lines 15-17). This reduction in sample size, associated with the higher frequency in the sampling process, intents to reduce the measuring overhead in the measurement point.

If $m > m_{max}$ the value of the reference parameter is changing slower than predicted, then the interval between samples can be increased, reducing the measurement impact. In this moment of less activity in the network, the sample size is increased by a factor of $1/K$, with $K = 10$ (lines 21-23). This allows to collect more information about the network in less critical periods of its operation.

If the value S does not change since the last sample $X[order]$, m assumes an undefined value, which indicates that the network is stable. In this case, the interval between sample is exponentially increased and the sample size is increased by a factor of $1/K$ (lines 24-26), with $K = 10$.

Additional constraints are used to prevent ΔT from growing indefinitely, thus maintaining a minimum number of samples to ensure data to new predictions. Conversely, the maximum sampling frequency is also limited, so that the sampling interval does not tend to zero, which would resulting in capturing all traffic. These limits were tuned experimentally and define the minimum interval between samples as 0.1s and the maximum interval between samples as 8s (lines 28-33). Similarly, restrictions are defined to limit the sample size variation. These restrictions avoid samples extremely small, which difficult estimating parameters statistically, as well as samples excessively large, closely matching a capture of total traffic. These limits were set experimentally, determining the minimum sample size as 0.1s and the maximum sample size as 2s (lines 35-40). The new interval between samples and the sample size are then used in the next capture (lines 41-42). The values of ΔT_{actual} and S are stored in the vectors T and X (line 44) to be used in the next iteration of the multiadaptive technique.

IV. Proof-of-concept

The proof-of-concept aims: (i) to assess the multiadaptive sampling strategy (MA) ability to capture the network behavior correctly with reduced overhead; (ii) to compare the performance gain of the strategy facing currently used techniques; and (iii) to demonstrate the versatility of the proposed strategy, evaluating its effectiveness in distinct traffic scenarios.

For this purpose, a statistical and visual analysis of the reference parameter evaluated using the several sampling techniques is cross-checked with the values obtained using the total traffic. Although the statistical parameters in use are common in representative research on adaptive sampling, this work extends previous works by cross-checking results from sampling against the total traffic. The common evaluation approaches only compare the performance of the proposed techniques to the Systematic Time-based technique [2].

The reference parameter adopted in the tests is network throughput, which is a proper indicator about the network activity variation. The order of prediction N is set to three.

A. Evaluation Scenarios

Table I illustrates the characteristics of the real traffic traces used in the tests.[1]

TABLE I
Traffic Scenarios

Traffic Label	Characteristics	Duration	Available
SIGCOMM08	Captured during the SIGCOMM 2008 conference, including all the participants communications through IEEE 802.11a access points.	8 hours	CRAWDAD [16]
OC-48	Captured passively in a backbone link OC - Optical Carrier 48 in a large ISP from the US West Coast	5 min.	CAIDA [17]
SIP VoIP	Capture of the VoIP traffic, using SIP and G.711 encoding, in a Brazilian university campus with Fast Ethernet links	20 hours	University of Sergipe, Brazil
Video streaming	Capture of a live stream transmission of the STS-135 launching mission conducted by NASA on July 8, 2011. The stream was broadcast in High Definition 720p, through NASA TV channel and encoded in MPEG-4 using RTP as transport protocol.	15 min.	University of Minho, Portugal

To assess the achieved optimization levels, the MA technique is compared to the Systematic Time-based technique [2] and Adaptive Linear Prediction technique [13]. *Systematic Time-based Sampling (ST)* is one of the most common sampling technique currently used. In the sampling process the packet selection follows a deterministic function based on the arrival time to the measurement point, i.e., the sample size and the time between samples are set at the beginning and are unchanged along the sampling process. In this work, as suggested in [13], the operational parameters sample size and time between samples have been set to 0.1s and 0.5s, respectively. *Adaptive Linear Prediction Sampling (LP)* adjusts the time interval between samples based on the level of network activity using linear prediction. However the sample size is fixed along the sampling process. Here, the initial interval between samples was set to 0.2s and the sample size was set to 0.1s. Another configurable parameter of this technique is the order of prediction, which has been set as defined in the original specification [13] and corresponds to 2. Although the technique takes into account the last two samples to decide the next interval between samples, the measurement point still needs to maintain information about all traffic since the beginning of its execution. This is needed because the technique analyses the evolution of the reference parameter based on the accumulated amount of data until the end of the sample N, relative to the sample $N-1$.

B. Statistical analysis

The main goal of the statistical analysis is to determine the equivalence between the total traffic behavior and the behavior estimated through the sampling process, evaluating simultaneously the overhead associated with the amount of packets selected by the measurement technique. Additionally, it intents to quantify the efficiency of the proposed strategy facing other proposals. The most common way to analyse the traffic behavior is through the instantaneous throughput, measuring the amount of data traffic in a time interval, usually in bps. This metric is commonly used for a graphical and statistical representation of the network activity in real time.

The above goals are verified by comparing the techniques through the parameters presented on table II.

V. Evaluation Results

A. Overhead reduction

Table III quantify the overhead of each sampling technique under evaluation considering the overhead metrics defined in Table II. As shown, the multiadaptive technique leads to a significant overhead reduction for all traffic types.

As an example, for OC-48 traffic, the sampled traffic resulting from the multiadaptive technique corresponds to 5.14% of total data in the original trace. Comparing to the systematic technique, the multiadaptive technique reduces in 68.87% the number of collected packets, in 68.94% the total amount of data sampled, and in 73.40% the number of samples used to characterize the OC-48 traffic. For the same traffic type, regarding the adaptive LP technique, the multiadaptive technique achieves a reduction of 63.58% in the number of collected packets, 63.58% in the total data amount, and

[1] Although evaluation tests are based on traffic traces previously captured, the adaptive parameters of the sampling techniques are configured dynamically during the sampling process. Similarly to a real-time operational environment, there are no previous knowledge of the characteristics of the traffic passing through.

TABLE II
STATISTICAL PARAMETERS

Overhead	**Measurement Goal:** Evaluate the overhead associated with the amount of packets processed, stored and transmitted by the measurement point where the sampling technique is deployed.		
	Metrics		
	Number of Packets	Total number of packets captured during the sampling process for each sampling technique.	#packets
	Amount of Data	Sum of all packets collected with each sampling technique. For this metric, it is used the total length field within IP header.	Mbyte
	Number of Samples	Total number of samples captured during the sampling process.	#samples
Throughput	**Measurement Goal:** Evaluate the accuracy of the sampling technique in estimating network throughput.		
	Metrics		
	Throughput	Measure the amount of data traffic in a time interval.	kbps
	Coefficient of Variation	Measure the variability of time series. In network measurements, helps to identify and characterize the burstiness of traffic.	Ratio between standard deviation and average throughput.
	Explosivity	Complementary descriptive statistics to measure the variability of time series.	Ratio between the peak and average throughput in a measurement interval.
	Correlation	Assess the correlation between the sampled traffic and total traffic, studying statistically the relationship between the corresponding series.	Coefficient of correlation ρ (Pearson method [18]): $0.7 < \rho \leq 1$, strong correlation; $0.3 < \rho \leq 0.7$, moderate correlation; $0 \leq \rho \leq 0.3$, weak correlation
	Relative Mean Error	Assess the discrepancy between the mean of the total traffic and its sampled version.	$$RME = \frac{\lvert M_{total} - M_{estimated} \rvert}{M_{total}} \quad (1)$$ M_{total} is the average throughput of total traffic; $M_{estimated}$ is the average throughput of the sampled traffic [7].

69.14% in the number of processed samples. As shown in Table III, for the remaining traffic types the overhead reduction achieved is similar.

These results clearly attest the performance improvement for applying the proposed technique in measurement points. However, to prove measurement efficiency the statistical representativeness of the sampled traffic has to be evaluated, verifying its ability to capture real traffic behavior. This evaluation is provided below.

TABLE III
OVERHEAD REDUCTION FOR ALL TRAFFIC TYPES

Traffic / Parameter	Total	ST	LP	MA
OC-48				
Number of packets	6550395	1094002	935066	340577
Data volume (MBytes)	3189.67	533.39	454.92	165.72
Number of samples		500	431	133
SIP VoIP				
Number of packets	1172378	188471	190572	62650
Data volume (MBytes)	168.68	27.41	32.27	9.03
Number of samples		105385	53081	27176
SIGCOMM08				
Number of packets	4513615	749977	899145	226060
Data volume (MBytes)	2382.92	395.97	511.93	120.49
Number of samples		44511	37104	9598
Video streaming				
Number of packets	97248	16303	13705	4711
Data volume (MBytes)	101.76	17.02	14.86	4.85
Number of samples		1090	406	362

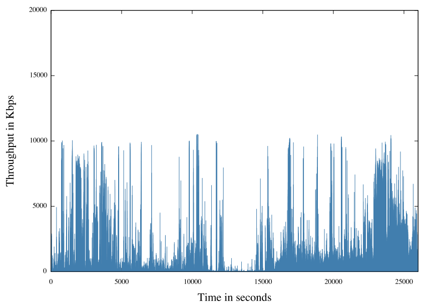
B. Throughput estimation

To describe traffic behavior, a visual representation of network throughput is commonly available in monitoring tools, illustrating the network workload at each measurement interval. Figure 1 presents the instantaneous throughput measured in 1s time intervals for SIGCOMM08 traffic. As depicted, the multiadaptive sampling technique represents closely the total traffic behavior when visually compared.

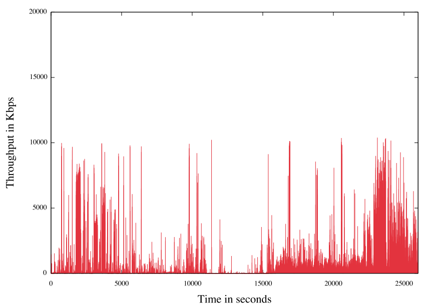
As regards throughput estimation, Table IV presents the statistical comparison resulting from applying each sampling technique for all traffic traces. The results show that systematic and multiadaptive techniques achieve high-quality representations of traffic behavior when comparing average throughput, explosivity, CV and correlation with the corresponding total traffic statistics. The adaptive LP technique tends to overestimate throughput when using low data rate traffic traces (e.g. VoIP traffic). The best match of this technique was achieved for OC-48 trace, which represents a high throughput link of a ISP. The correlation analysis of the instantaneous throughput time series (considering a measurement interval of 1s), obtained comparing each sampling strategy to the original trace, presents high coefficients of correlation, corroborating statistically the visual tendencies illustrated in Figure 1. The accuracy achieved by the multiadaptive technique is ratified when considering the relative mean error of the estimated average throughput (see Table IV).

For a more detailed analysis, Figure 2 presents the relative error spread for each sampling technique when applied to the SIGCOMM08 trace. As shown, for all the tested techniques the error distribution is similar, concentrated near to zero and with a well-bounded tail. The figure also shows that the higher relative errors result from overestimation, expressed by the positive tail in the histogram.

A complementary analysis of the coefficient of correlation and relative error can be obtained through the analysis of dispersion, represented in Figure 3, for SIGCOMM08 trace. These graphs illustrate the accuracy obtained in the instanta-



(a) Total traffic



(b) Multiadaptive sampling

Fig. 1. Throughput: SIGCOMM08

TABLE IV
OVERALL ESTIMATED BEHAVIOR

Traffic / Parameter	Total	ST	LP	MA
SIGCOMM08				
Throughput	730.95	728.76	1130.27	803.65
RME		0.002	0.54	0.09
Explosivity	14.39	15.16	10.05	12.93
C.V.	1.90	2.07	1.56	1.93
Correlation		0.91	0.91	0.89
SIP VoIP				
Throughput	21.85	21.30	50.49	22.03
RME		0.02	1.31	0.008
Explosivity	158.40	132.14	58.00	112.72
C.V.	5.28	5.41	2.90	5.08
Correlation		0.71	0.88	0.86
OC-48				
Throughput	87103.35	87390.26	86467.86	86825.99
RME		0.003	0.007	0.003
Explosivity	1.40	1.35	1.34	1.23
C.V.	0.10	0.09	0.10	0.13
Correlation		0.84	0.81	0.82
Video streaming				
Throughput	1275.84	1279.17	2998.53	1425.09
RME		0.002	1.35	0.11
Explosivity	19.13	19.98	8.41	17.02
C.V.	3.93	3.96	2.50	3.66
Correlation		0.99	0.98	0.98

neous throughput estimation, i.e., considering the throughput evaluated in each measuring time interval, where at least a single packet is sampled. The deviation evinces the difference between the estimated instantaneous throughput and the real instantaneous throughput. A concentration of dots near the reference line indicates a close match between these measures. As

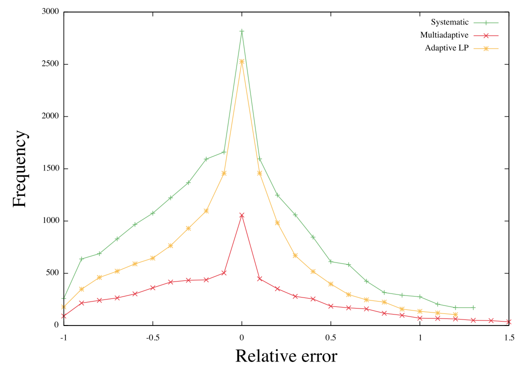
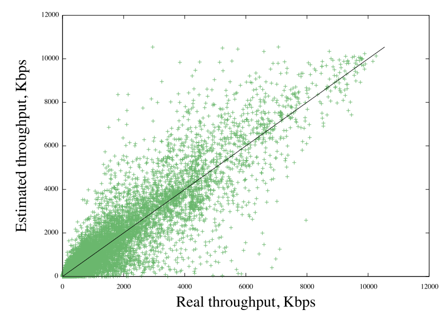
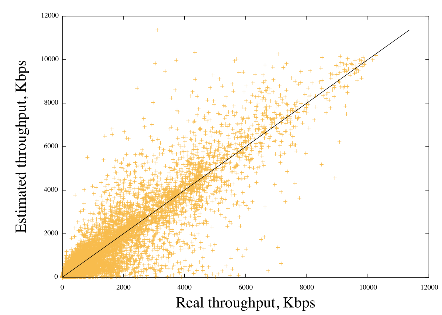


Fig. 2. Error frequency: SIGCOMM08

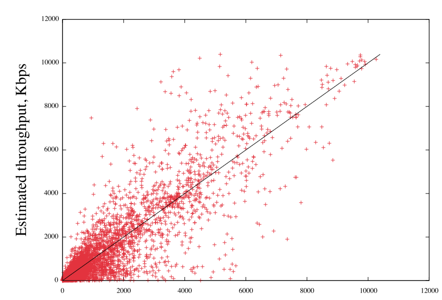
shown, all the sampling techniques exhibit a higher dot density for low rates and a similar visual spectrum, corroborating the relative error results presented in Figure 2.



(a) Systematic



(b) Adaptive LP



(c) Multiadaptive

Fig. 3. Estimative spread: SIGCOMM08

Regarding the multiadaptive technique, when analysing the trade-off between accuracy and overhead reduction in Tables IV and III, it is clear that this technique promotes a significant improvement comparing to the other sampling techniques considered. These results show that, despite the significant reduction on the traffic volume considered, the multiadaptive technique ability to capture the real traffic behavior correctly is not compromised. For several statistical parameters, the proposed technique outperforms the conventional techniques, which is even more relevant attending to the significant decrease on measurement overhead.

C. Studying the impact of the order of prediction

These experiments aim at evaluating the impact of varying the order of prediction N on the performance of the MA technique. Table V presents the results comparing overhead and throughput estimation for all traffic types and four distinct values of N. Notice that higher values of N correspond to configuring the adaptive traffic selection process with more information of past samples, i.e., including more past memory. Generally, a larger past memory leads to less reactive traffic mechanisms, and shorter past memory improves the reactivity to short term traffic fluctuations. The degree of this reactivity may affect the stability of traffic mechanisms.

According to the obtained results, an increase in N conducts to an overhead reduction regarding the number of collected samples, for all traffic types. However, for $N = 5$, it is clear that the overhead reduction progression is not always verified when taking the number of packets and data volume as metrics. This is due to an increase in sample size, which is inline with the adaptive behavior of MA technique mentioned in Section III-B. Regarding RME, increasing the order of prediction may also increase the estimation error. Nevertheless, when taking $N = 2$, the high overhead does not necessary imply a gain in accuracy.

As mentioned previously, in the results from previous sections the order of prediction N was set to three, which represents a good compromise between overhead and accuracy. Considering the heaviest traffic trace (OC-48) the results could be optimized considering $N = 4$ as the evident overhead reduction does not penalize RME.

As a whole, see Table V, the values obtained for RME are very low in all cases. Therefore, depending on the objective or usefulness of throughput estimation, the value of N can be tuned to achieve an adequate compromise between overhead and efficiency.

TABLE V
IMPACT OF ORDER OF PREDICTION N ON MA PERFORMANCE

Traffic / Parameter	Total	$N = 2$	$N = 3$	$N = 4$	$N = 5$
OC-48					
Number of packets	6550395	366229	340577	286403	302894
Data volume	3189.67	178.37	165.72	139.36	147.47
Number of samples		151	133	105	93
Throughput	87103.4	86202.3	86825.9	86818.8	86339.6
RME		0.01	0.003	0.003	0.008
VoIP					
Number of packets	1172378	81213	62650	57539	66183
Data volume	168.68	11.11	9.03	8.06	9.24
Number os samples		42277	27176	19636	16777
Throughput	21.85	18.55	22.03	22.19	24.02
RME		0.15	0.008	0.01	0.09
SIGCOMM08					
Number of packets	4513615	240159	226060	222466	267925
Data volume	2382.92	128.78	120.49	120.99	146.61
Number of samples		12152	9598	8324	7259
Throughput	730.95	732.00	803.65	819.92	809.05
RME		0.001	0.09	0.12	0.10
Video Streaming					
Number of packets	97248	4327	4711	4207	3362
Data volume	101.76	4.22	4.85	4.41	3.50
Number of samples		362	239	187	183
Throughput	1273.84	815.49	1425.09	1561.17	1088.91
RME		0.35	0.11	0.22	0.14

D. Studying the impact of the activity period

According to [9], adaptive sampling techniques tend to be less accurate when the network activity is low. The tests included in this section intent to assess the MA technique ability in estimating network throughput correctly irrespectively of the network load period.

When observing Figure 1 it is clear that the time period between seconds 12500 and 14000 corresponds to a low activity period; conversely, the period between seconds 23000 and 25000 corresponds to a peak period in users' activity. Therefore, the performance of each sampling technique was evaluated for these two distinct periods, to verify its versatility in estimating network behavior despite the network activity.

TABLE VI
IMPACT OF NETWORK ACTIVITY - SIGCOMM08

Low network load				
Parameter	**Total traffic**	**ST**	**LP**	**MA**
Number of packets	5291	592	257	464
Data volume (MBytes)	1.19	0.15	0.05	0.11
Number of samples		2467	904	499
Throughput (kbps)	236.03	266.33	4.96	271.13
RME		0.12	0.97	0.14
High network load				
Parameter	**Total traffic**	**ST**	**LP**	**MA**
Number of packets	901589	149634	226401	44500
Data volume (MBytes)	661.90	110.09	166.02	32.66
Number os samples		3333	5183	747
Throughput (kbps)	769.81	771.46	1624.14	769.74
RME		0.002	1.1	0.00009

The results presented in Table VI corroborate the results obtained in [9] as the LP technique clearly under-estimates network throughput for low activity periods. Moreover, this technique is also inaccurate for high activity periods, over-estimating throughput. Irrespectively the activity period, ST and MA techniques obtained good results considering the estimation accuracy. However, the efficiency of MA technique is higher, as the estimation process involves less overhead, when taking all the related metrics (number of samples, number of packets and traffic volume). For high activity periods the accuracy-overhead trade-off was even more encouraging.

Table VII explores the impact of the order of prediction in

TABLE VII
ANALYSIS BY ORDER - SIGCOMM08

Low network load					
Traffic / Parameter	**Total**	$N = 2$	$N = 3$	$N = 4$	$N = 5$
Number of packets	5291	456	464	486	953
Data volume	1.19	0.13	0.11	0.10	0.17
Number of samples		863	499	417	332
Throughput	236.03	301.74	271.13	235.53	196.40
RME		0.27	0.14	0.002	0.16
High network load					
Number of packets	901589	44886	44500	38846	38141
Data volume	661.90	33.02	32.66	28.69	27.91
Number of samples		810	747	641	553
Throughput	769.81	771.58	769.74	774.61	767.57
RME		0.002	0.00009	0.002	0.16

the obtained results. While for low activity periods $N = 4$ is the best choice, for high activity periods the order of prediction can be higher to reduce overhead without compromising MA efficiency.

The above results evince that the proposed MA technique is a step forward regarding classic adaptive sampling techniques, being simultaneously more accurate and versatile.

VI. CONCLUSIONS

This paper has presented a self-adaptive traffic sampling technique aiming at optimizing the usual trade-off between network estimation overhead and accuracy. By changing both the interval between samples and the sample size according to the observed network activity, this technique was able to capture correctly the network throughput with very low overhead. The performance of this technique was evaluated using the Systematic Time-based and Adaptive Linear Prediction as benchmarking techniques. Using real traffic representing distinct profiles, this paper provides a comparative statistical analysis of measurement overhead and estimation accuracy, under distinct traffic load situations. This analysis has demonstrated the effectiveness and versatility of the proposal, outperforming the conventional techniques. To optimize sampling efficiency, the impact of the order of prediction on the results for the distinct traffic situations was also evaluated.

The proposed technique is simple and easy to deploy in network measurement points. Presently a multiadaptive sampling framework is under development aiming to facilitate the technique usage on operational scenarios.

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